American International University-Bangladesh (AIUB)

Department of Computer Science (Spring 2025-26)

Assignment 1, HCI Theory

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Part (A): Design Issues in Everyday Things (Affordance Analysis)

Object : Escalator

1. Visibility

- **Problem Situation :** When an escalator is stopped, a user approaching the escalator may hesitate because the cannot determine the direction.
- **Affordance Violation :** The direction indicators or arrows are not clearly visible.
- **Suggested Improvement :** Large LED arrows, bright direction signs, or floor indicators could clearly show the direction.
- **Relation to Ubiquitous Computing :** According to Mark Weiser, technology should be easy and natural to use without requiring conscious effort. Clear indicators would make the interaction effortless.

2. Mapping

- **Problem Situation :** Sometimes the physical layout of escalators does not clearly indicate which one goes up and which one goes down.
- **Affordance Violation :** The mapping between the users action (entering the escalator) and the systems function (movement direction) is not obvious.
- **Suggested Improvement :** Directional arrows on the floor can improve natural mapping.
- **Relation to Ubiquitous Computing :** Technology should provide natural relationships between controls and actions so users do not need to think about them.

3. System Image

- **Problem Situation :** When the escalator is not moving, the interface does not clearly communicate how the system works. Users may assume the escalator is simply a staircase rather than a machine that moves in a specific direction.
- **Affordance Violation :** The system image fails to provide enough information about the escalators function.
- **Suggested Improvement :** Direction lights, digital displays, or illuminated arrows could clarify the systems behavior.
- **Relation to Ubiquitous Computing :** Technology should communicate its functionality clearly through its design.

4. Gulf of Execution

- **Problem Situation :** A person wants to go up but cannot immediately identify the correct escalator.
- **Affordance Violation :** The lack of visible directional cues fails to communicate how the escalator should be used.
- **Suggested Improvement :** Clear UP and DOWN signs, floor markings, or illuminated arrow indicators should be placed at the entrance of the escalator.
- **Relation to Ubiquitous Computing :** According to Mark Weiser, technology should minimize the gap between a users intention and the required action. Clear directional indicators would allow users to interact with the escalator effortlessly.

5. Gulf of Evaluation

- **Problem Situation :** A user stands on the escalator and only realizes after a few seconds that it is moving downward instead of upward.
- **Affordance Violation :** The escalator does not provide clear feedback about its operational state or direction before it starts moving. This lack of feedback prevents users from easily evaluating whether their action was correct.
- **Suggested Improvement :** Moving arrow lights, digital direction indicators, or audible signals that activate before the escalator starts moving.
- **Relation to Ubiquitous Computing :** In the vision of Mark Weiser, technology should provide clear feedback so users can quickly understand the systems current state without confusion.

Part (B): Human Errors (Slips)

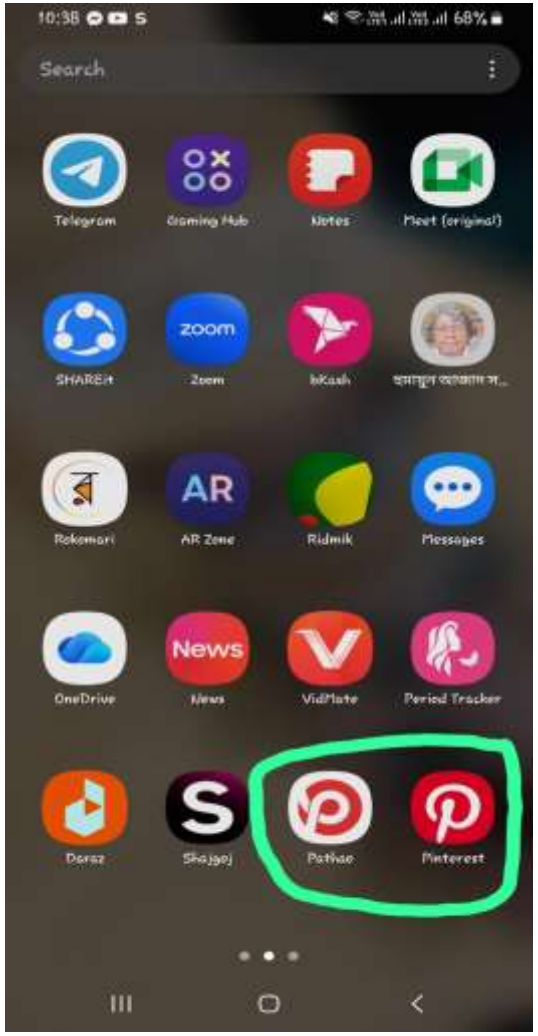
Slip 1 : Opening the wrong app.

Situation : I intended to open the **Pathao** app to book a ride, but accidentally tapped the **Pinterest** app because their icons looked somewhat similar and were placed close together on my home screen.

Type of Slip: Attention Slip.

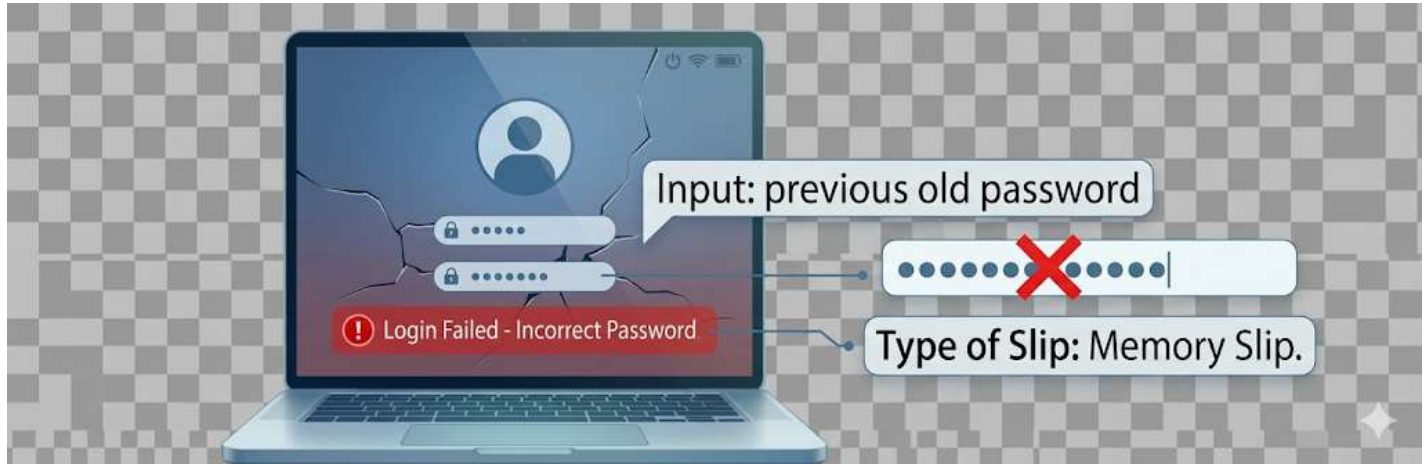
Design Prevention: App icons could use more distinctive colors, shapes, or spacing to reduce confusion.

Weiser Vision: According to Mark Weiser's calm computing vision, technology should assist users quietly without demanding attention. If app icons were more visually distinct or better organized, the system could subtly prevent such mistakes, allowing users to interact smoothly without confusion.



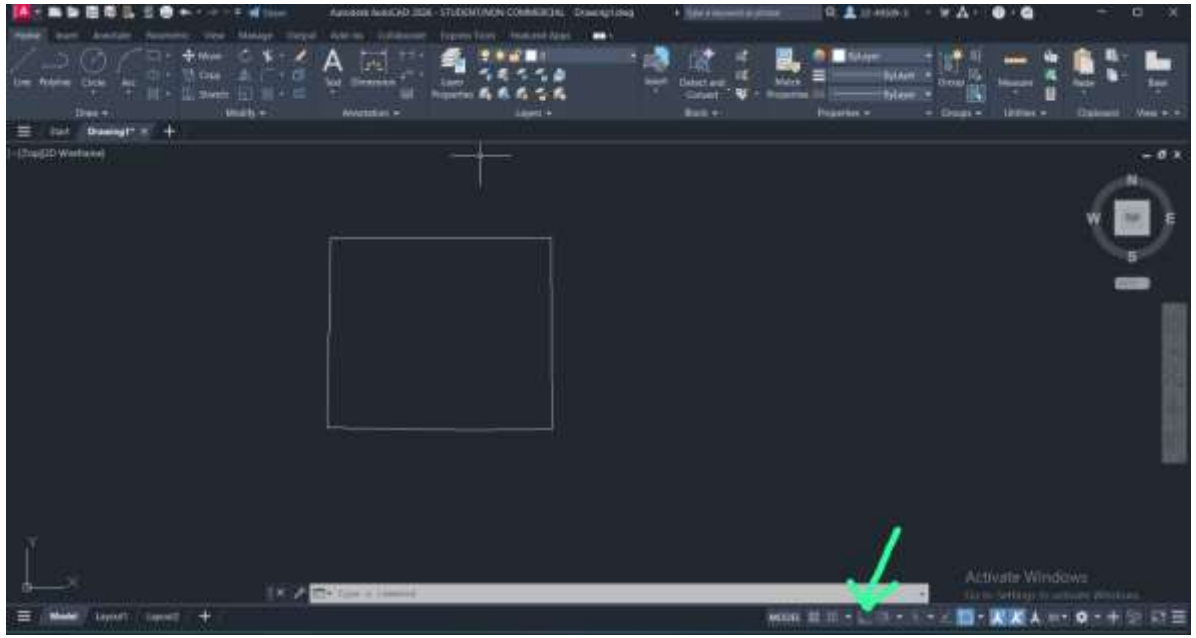
Slip 2 : Laptop login wrong password

- **Situation:** I tried to log in to my laptop but typed the previous old password.
- **Type of Slip:** Memory Slip.
- **Design Prevention:** Provide a Hint option on the login screen so users can recall the correct password without stress.
- **Weiser Vision:** The Hint works invisibly in the background, keeping the workflow smooth and aligning with Mark Weiser's concept.



Slip 3: Drawing a Straight Line in AutoCAD without Ortho Mode On

- **Situation:** I tried to draw a perfectly horizontal line in AutoCAD, but I forgot to turn Ortho Mode ON. As a result, the line was slightly diagonal instead of straight.
- **Type of Slip:** Mode Error.
- **Design Prevention:** Provide subtle visual cues or prompts if the user tries to draw a line that deviates from horizontal/vertical while Ortho Mode is OFF.
- **Weiser Vision** : visual guidance in the interface would have prevented the slip, making the design experience smoother, calmer, and more aligned with **calm computing principles**.



Slip 4 : Use Wrong Variable.

- **Situation:** While writing a program, I created two variables named totalMarks and totalMark. Because the names looked very similar, I accidentally used totalMark in the calculation instead of totalMarks, which caused the program to produce the wrong output.
- **Type of Slip:** Data-Driven Error.
- **Design Prevention:** Use clear and distinct variable names and Use code editors with syntax highlighting or auto complete to help identify the correct variable.
- **Weiser Vision** : Technology should support the user quietly, anticipating potential mistakes and helping prevent them without being intrusive. Using a smart editor or IDE with calm, context-aware features like syntax highlighting, auto-complete, and subtle warnings would have prevented this slip, making programming more seamless and less error-prone.

```cpp
#include <iostream>
using namespace std;

int main() {
    int math = 80;
    int physics = 85;
    int chemistry = 90;

    int totalMarks;
    int totalMark;

    totalMarks = math + physics + chemistry;
    totalMark = 0;

    cout << "Total Marks = " << totalMark << endl;

    return 0;
}
```

Output:

```
Total Marks = 0
```

Slip 5: Syntax Error.

- **Situation:** While writing a program, I usually use printf() in C language. When I started writing a C++ program, out of habit I typed printf() instead of cout , even though cout should be used in C++.
- **Type of Slip:** Associative Activation Error.
- **Design Prevention:** Use compiler warnings or suggestions that indicate when a more appropriate C++ method should be used.
- **Weiser Vision** : The slip could be prevented if the system provided calm, context-aware guidance, aligning perfectly with Weiser's vision of computing that helps users without demanding attention.

```
#include <iostream>
#include <stdio.h>
using namespace std;

int main() {
    int number = 10;

    printf("The number is %d\n", number);

    return 0;
}
```

Correct C++ Version:

```
#include <iostream>
using namespace std;

int main() {
    int number = 10;

    cout << "The number is " << number << endl;

    return 0;
}
```

Slip 6: Google Colab Context.

- **Situation :** I was training a machine learning model in Google Colab. I remembered that I needed to upload the dataset to my Google Drive first. I started writing the training code, but halfway through, I forgot to upload the dataset. When I ran the code, it failed with a File not found error. Only then did I remember that the dataset was not uploaded. I went back to my Drive, uploaded the dataset, and retried the training successfully.
- **Type of Slip:** Loss-of-Activation Error.
- **Design Prevention:** Google Colab could remind users to mount Drive and check for files before running the code.
- **Weiser Vision :** Weiser proposed that technology should support users quietly, without demanding constant attention or disrupting their workflow. This slip illustrates a lack of calm computing support a proactive, invisible nudge (like pre-run dataset verification) would have prevented the loss-of-activation error and made the computing experience more seamless.

References

- https://en.wikipedia.org/wiki/The_Design_of_Everyday_Things
- https://www.scribd.com/document/860698481/Artigo-Weiser-The-Computer-for-the-21st-Century-1